# Kernix 0.11.0 / Performance brief

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

**35** workloads

**119** observations

Source precision retained

## Matrix multiplication

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product




## How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

# Kernix 0.11.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

## Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · independent operands




## Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · independent operands



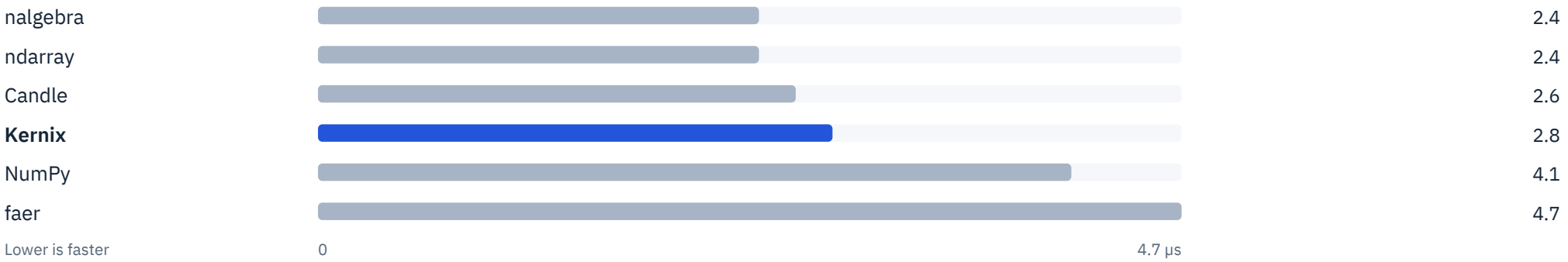
# Kernix 0.11.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

## Matrix multiplication

Per-call median

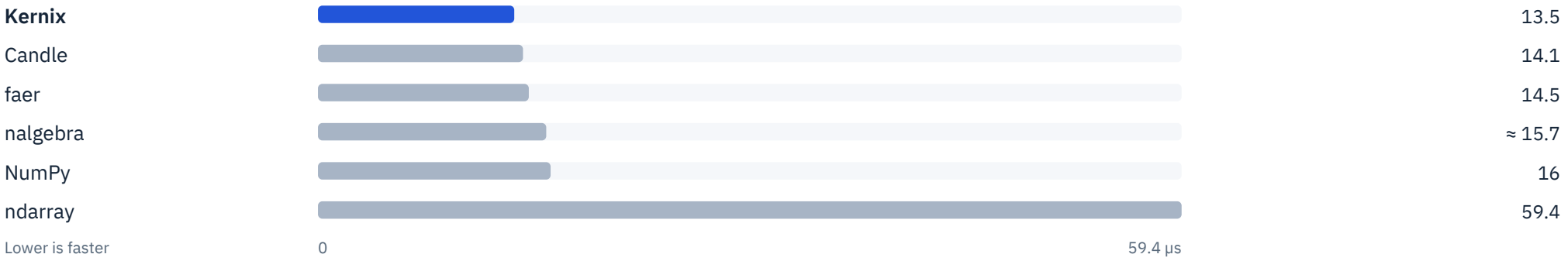
32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product




## Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



# Kernix 0.11.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

## Matrix multiplication

Per-call median

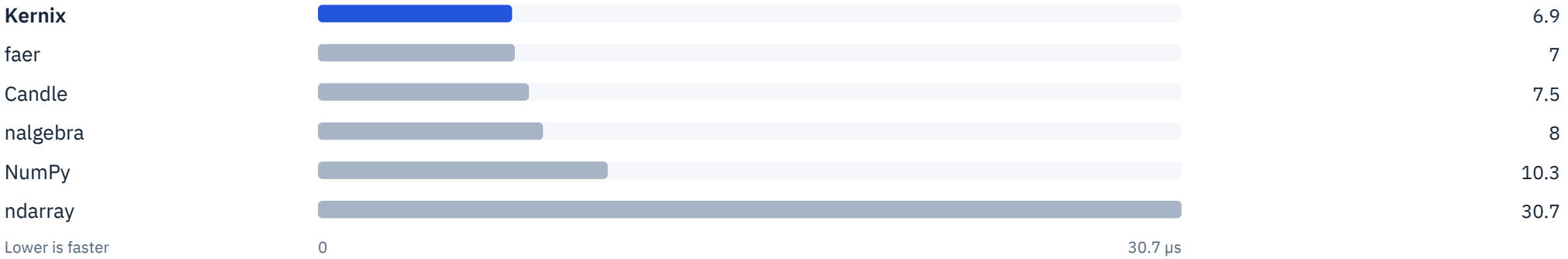
256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product




## Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



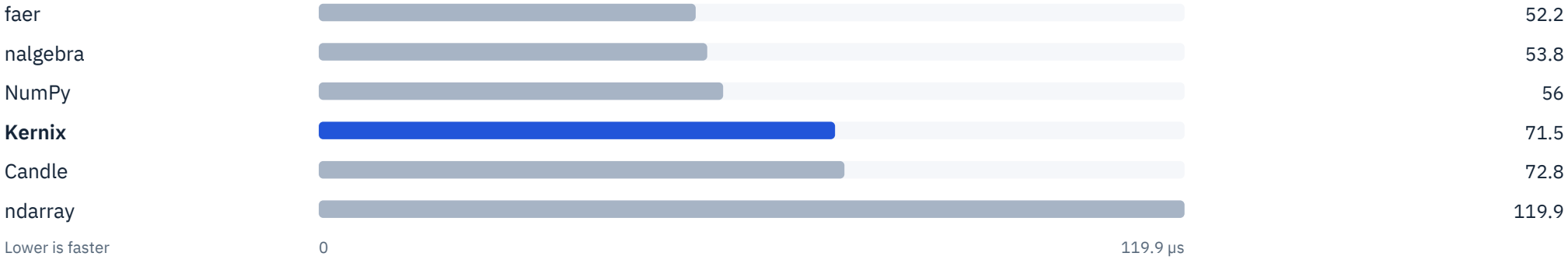
# Kernix 0.11.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

## Matrix multiplication

Per-call median

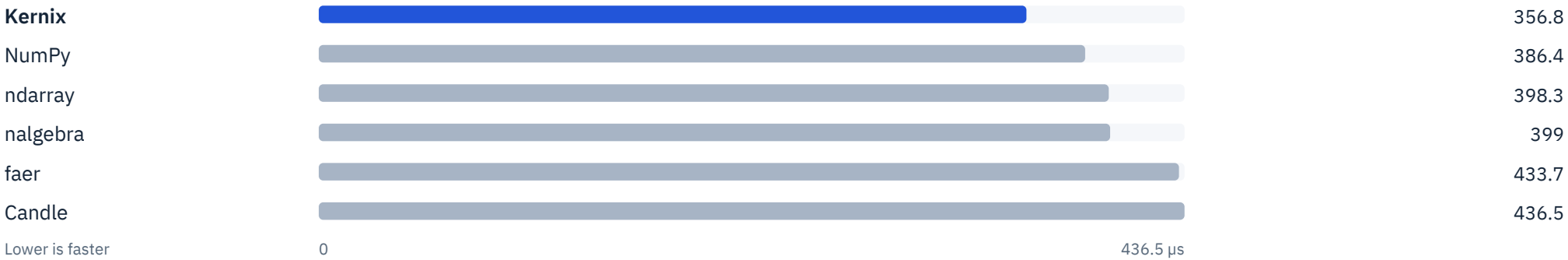
128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product




## Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product



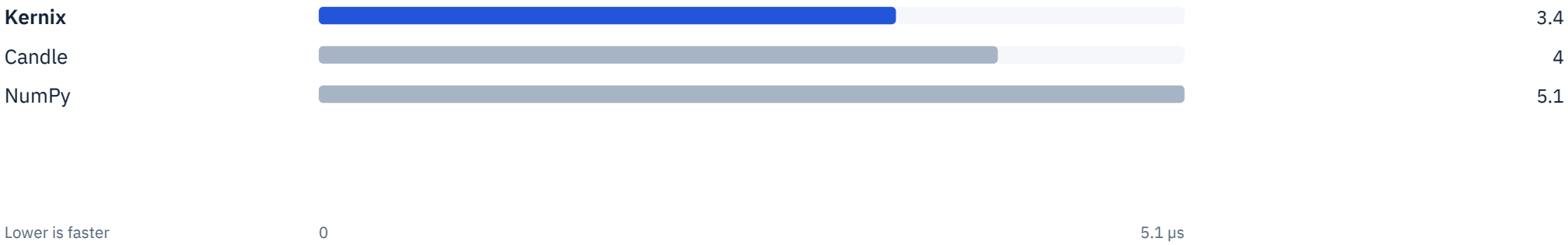
# Kernix 0.11.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

## Batched multiplication

Per-call median

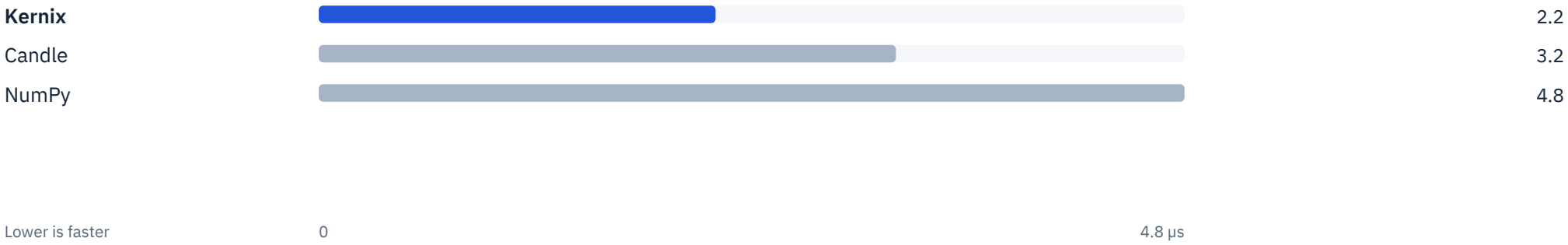
4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches




## Batched multiplication

Per-call median

4 × 16 × 32 · 1 × 32 × 16 · f32 · 1 worker · allocated output · broadcast batches



# Kernix 0.11.0 / Explicit parallel execution

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

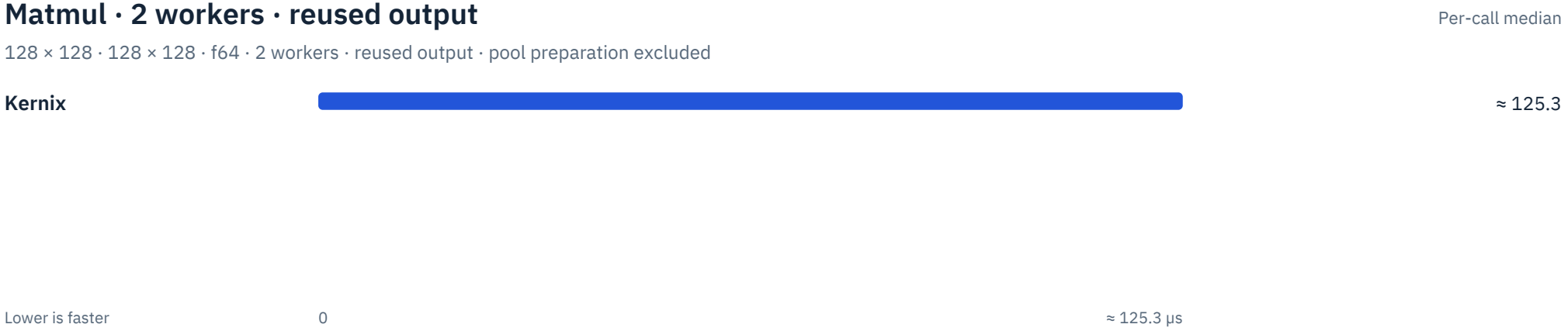
## Matmul · 2 workers · reused output

64 × 64 · 64 × 64 · f64 · 2 workers · reused output · pool preparation excluded




## Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f64 · 2 workers · reused output · pool preparation excluded



# Kernix 0.11.0 / Explicit parallel execution

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

## Matmul · 2 workers · reused output

Per-call median

256 × 256 · 256 × 256 · f64 · 2 workers · reused output · pool preparation excluded




## Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f32 · 2 workers · reused output · pool preparation excluded

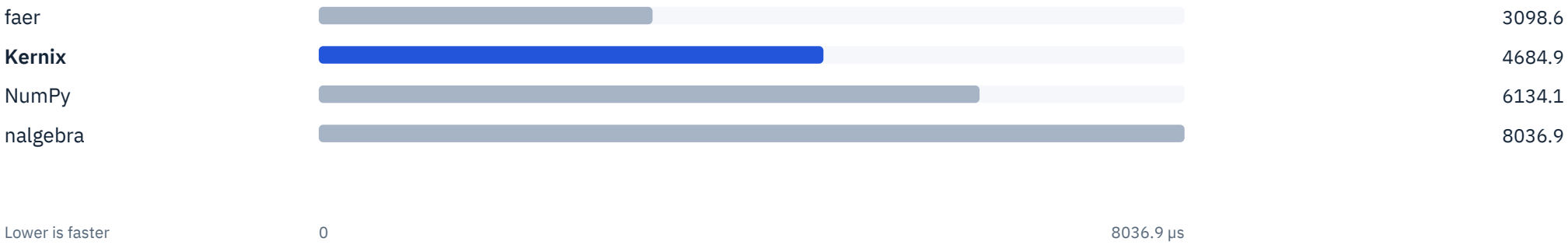


# Kernix 0.11.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

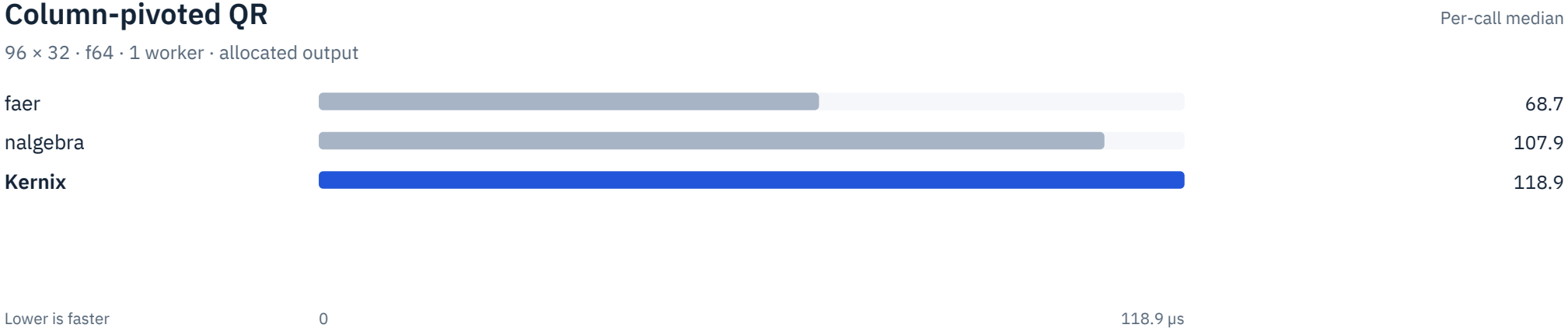
## Cholesky factorization

512 × 512 · f64 · 1 worker · allocated output




## Column-pivoted QR

96 × 32 · f64 · 1 worker · allocated output



# Kernix 0.11.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

## Column-pivoted QR

256 × 64 · f64 · 1 worker · allocated output

Per-call median




## Minimum-norm least squares

64 × 32, rank 16, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ

Per-call median



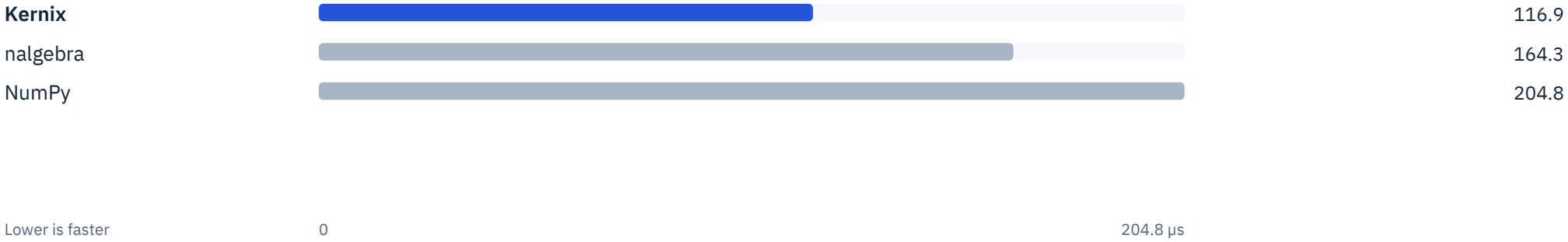
# Kernix 0.11.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

## Minimum-norm least squares

Per-call median

32 × 64, rank 32, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ




## Thin QR

Per-call median

96 × 32 · f64 · 1 worker · allocated output

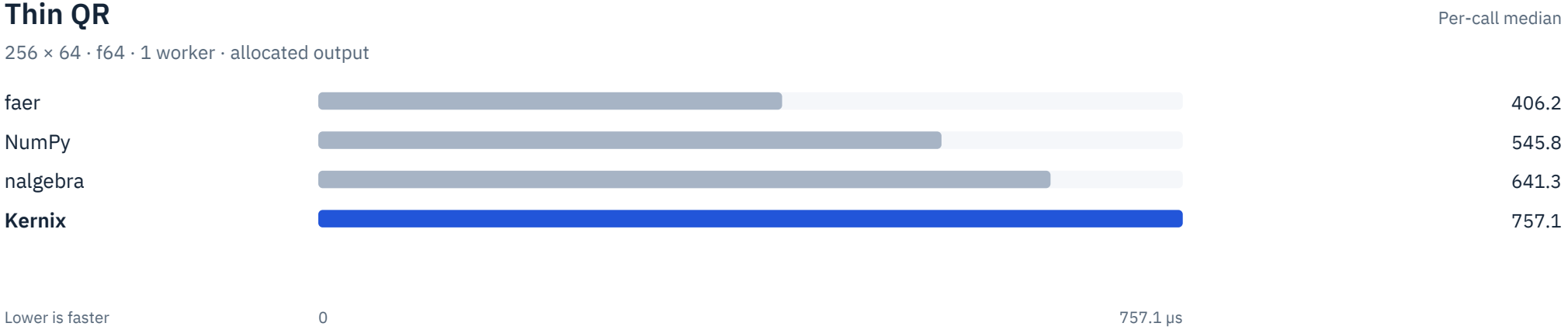


# Kernix 0.11.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

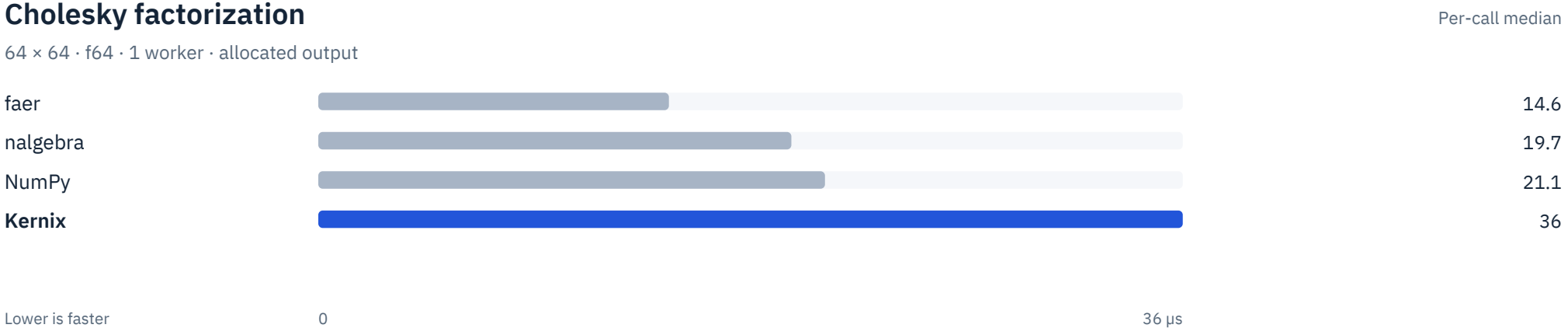
## Thin QR

256 × 64 · f64 · 1 worker · allocated output




## Cholesky factorization

64 × 64 · f64 · 1 worker · allocated output



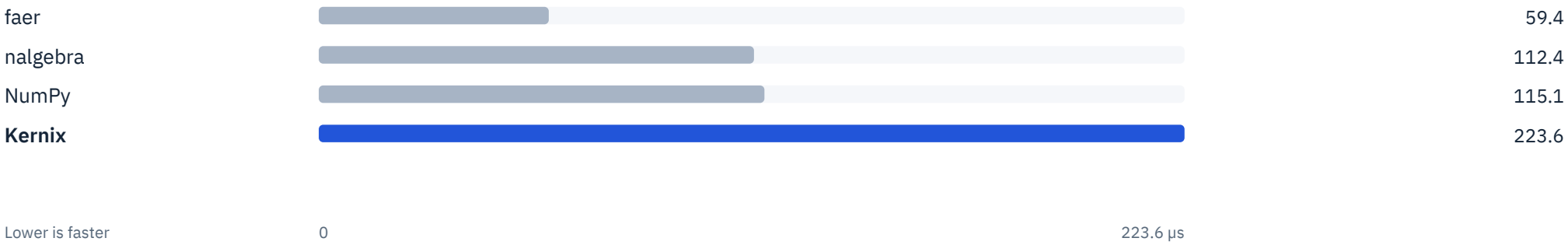
# Kernix 0.11.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

## Cholesky factorization

Per-call median

128 × 128 · f64 · 1 worker · allocated output




## Cholesky factorization

Per-call median

256 × 256 · f64 · 1 worker · allocated output



# Kernix 0.11.0 / Training & autodiff

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

## Shared-variable backward

Per-call median

sum(X * X), 64 × 32; prepared graph · f64 · 1 worker · prepared graph · fresh gradients




## Record + backward

Per-call median

sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



# Kernix 0.11.0 / Training & autodiff

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

## Model inference

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented




## Model recording + forward

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



# Kernix 0.11.0 / Training & autodiff

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

## Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients




## Parameter update

Per-call median

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented



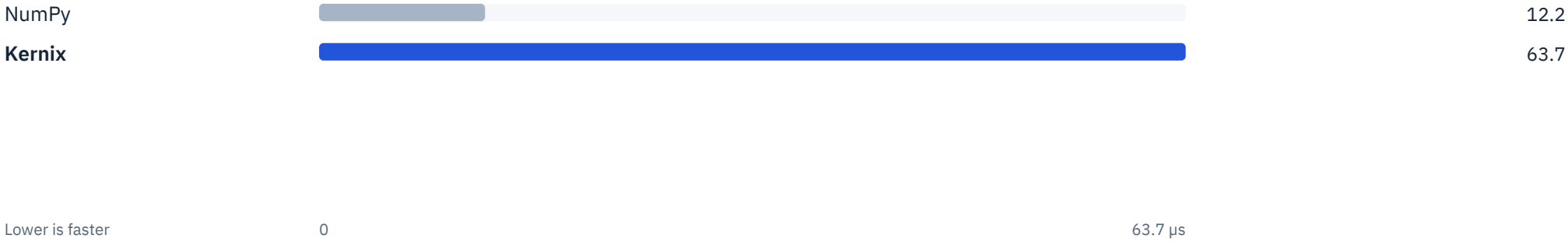
# Kernix 0.11.0 / Tensor contractions

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

## Coalesced tensor contraction

16 × 32 × 8 · 32 × 8 × 16 · f64 · 1 worker · allocated output

Per-call median

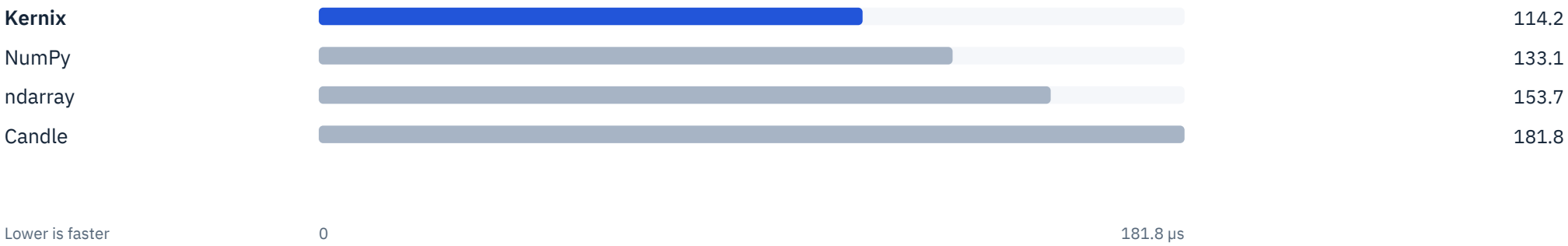


# Kernix 0.11.0 / Statistics & activations

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

## Row softmax

256 × 64 · f64 · 1 worker · allocated output



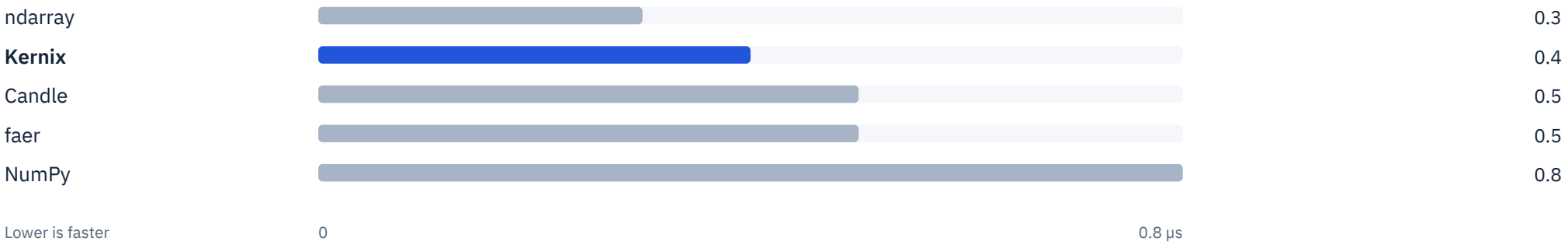
# Kernix 0.11.0 / Tensor construction

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

## Create zeros

Per-call median

64 × 64 · f32 · 1 worker · allocated output




## Create ones

Per-call median

64 × 64 · f32 · 1 worker · allocated output



# Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Independent matrix operands / NumPy**<br>128 × 128 · 128 × 128 · f64 | 101400 | 101.4 |
| **Independent matrix operands / Kernix**<br>128 × 128 · 128 × 128 · f64 | 140000 | 140 |
| **Independent matrix operands / NumPy**<br>128 × 128 · 128 × 128 · f32 | 56400 | 56.4 |
| **Independent matrix operands / Kernix**<br>128 × 128 · 128 × 128 · f32 | 71700 | 71.7 |
| **Matrix multiplication / Candle**<br>32 × 32 · 32 × 32 · f64 | 2600 | 2.6 |
| **Matrix multiplication / ndarray**<br>32 × 32 · 32 × 32 · f64 | 2400 | 2.4 |
| **Matrix multiplication / nalgebra**<br>32 × 32 · 32 × 32 · f64 | 2400 | 2.4 |
| **Matrix multiplication / faer**<br>32 × 32 · 32 × 32 · f64 | 4700 | 4.7 |
| **Matrix multiplication / NumPy**<br>32 × 32 · 32 × 32 · f64 | 4100 | 4.1 |
| **Matrix multiplication / Kernix**<br>32 × 32 · 32 × 32 · f64 | 2800 | 2.8 |

# Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Matrix multiplication / ndarray**<br>64 × 64 · 64 × 64 · f64 | 59400 | 59.4 |
| **Matrix multiplication / NumPy**<br>64 × 64 · 64 × 64 · f64 | 16000 | 16 |
| **Matrix multiplication / Candle**<br>64 × 64 · 64 × 64 · f64 | 14100 | 14.1 |
| **Matrix multiplication / faer**<br>64 × 64 · 64 × 64 · f64 | 14500 | 14.5 |
| **Matrix multiplication / nalgebra**<br>64 × 64 · 64 × 64 · f64 | 15699.999999999998 | 15.699999999999998 |
| **Matrix multiplication / Kernix**<br>64 × 64 · 64 × 64 · f64 | 13500 | 13.5 |
| **Matrix multiplication / NumPy**<br>128 × 128 · 128 × 128 · f64 | 101300 | 101.3 |
| **Matrix multiplication / ndarray**<br>128 × 128 · 128 × 128 · f64 | 114300 | 114.3 |
| **Matrix multiplication / Kernix**<br>128 × 128 · 128 × 128 · f64 | 147500 | 147.5 |
| **Matrix multiplication / faer**<br>128 × 128 · 128 × 128 · f64 | 101300 | 101.3 |

# Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Matrix multiplication / nalgebra**<br>128 × 128 · 128 × 128 · f64 | 115900 | 115.9 |
| **Matrix multiplication / Candle**<br>128 × 128 · 128 × 128 · f64 | 137600 | 137.6 |
| **Matrix multiplication / NumPy**<br>256 × 256 · 256 × 256 · f64 | 773900 | 773.9 |
| **Matrix multiplication / Candle**<br>256 × 256 · 256 × 256 · f64 | 869700 | 869.7 |
| **Matrix multiplication / nalgebra**<br>256 × 256 · 256 × 256 · f64 | 907500 | 907.5 |
| **Matrix multiplication / ndarray**<br>256 × 256 · 256 × 256 · f64 | 1151500 | 1151.5 |
| **Matrix multiplication / faer**<br>256 × 256 · 256 × 256 · f64 | 1092100 | 1092.1 |
| **Matrix multiplication / Kernix**<br>256 × 256 · 256 × 256 · f64 | 855400 | 855.4 |
| **Matrix multiplication / ndarray**<br>64 × 64 · 64 × 64 · f32 | 30700 | 30.7 |
| **Matrix multiplication / NumPy**<br>64 × 64 · 64 × 64 · f32 | 10300 | 10.3 |

# Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Matrix multiplication / faer**<br>64 × 64 · 64 × 64 · f32 | 7000 | 7 |
| **Matrix multiplication / Kernix**<br>64 × 64 · 64 × 64 · f32 | 6900 | 6.9 |
| **Matrix multiplication / Candle**<br>64 × 64 · 64 × 64 · f32 | 7500 | 7.5 |
| **Matrix multiplication / nalgebra**<br>64 × 64 · 64 × 64 · f32 | 8000 | 8 |
| **Matrix multiplication / nalgebra**<br>128 × 128 · 128 × 128 · f32 | 53800 | 53.8 |
| **Matrix multiplication / faer**<br>128 × 128 · 128 × 128 · f32 | 52200 | 52.2 |
| **Matrix multiplication / Candle**<br>128 × 128 · 128 × 128 · f32 | 72800 | 72.8 |
| **Matrix multiplication / ndarray**<br>128 × 128 · 128 × 128 · f32 | 119900 | 119.9 |
| **Matrix multiplication / Kernix**<br>128 × 128 · 128 × 128 · f32 | 71500 | 71.5 |
| **Matrix multiplication / NumPy**<br>128 × 128 · 128 × 128 · f32 | 56000 | 56 |

# Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Matrix multiplication / nalgebra**<br>256 × 256 · 256 × 256 · f32 | 399000 | 399 |
| **Matrix multiplication / NumPy**<br>256 × 256 · 256 × 256 · f32 | 386400 | 386.4 |
| **Matrix multiplication / ndarray**<br>256 × 256 · 256 × 256 · f32 | 398300 | 398.3 |
| **Matrix multiplication / faer**<br>256 × 256 · 256 × 256 · f32 | 433700 | 433.7 |
| **Matrix multiplication / Kernix**<br>256 × 256 · 256 × 256 · f32 | 356800 | 356.8 |
| **Matrix multiplication / Candle**<br>256 × 256 · 256 × 256 · f32 | 436500 | 436.5 |
| **Batched multiplication / Candle**<br>4 × 16 × 32 · 1 × 32 × 16 · f64 | 4000 | 4 |
| **Batched multiplication / NumPy**<br>4 × 16 × 32 · 1 × 32 × 16 · f64 | 5100 | 5.1 |
| **Batched multiplication / Kernix**<br>4 × 16 × 32 · 1 × 32 × 16 · f64 | 3400 | 3.4 |
| **Batched multiplication / Candle**<br>4 × 16 × 32 · 1 × 32 × 16 · f32 | 3200 | 3.2 |

# Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Batched multiplication / NumPy**<br>4 × 16 × 32 · 1 × 32 × 16 · f32 | 4800 | 4.8 |
| **Batched multiplication / Kernix**<br>4 × 16 × 32 · 1 × 32 × 16 · f32 | 2200 | 2.2 |
| **Matmul · 2 workers · reused output / Kernix**<br>64 × 64 · 64 × 64 · f64 | 46500 | 46.5 |
| **Matmul · 2 workers · reused output / Kernix**<br>128 × 128 · 128 × 128 · f64 | 125300.00000000001 | 125.30000000000001 |
| **Matmul · 2 workers · reused output / Kernix**<br>256 × 256 · 256 × 256 · f64 | 592800 | 592.8 |
| **Matmul · 2 workers · reused output / Kernix**<br>128 × 128 · 128 × 128 · f32 | 57400 | 57.4 |
| **Cholesky factorization / nalgebra**<br>512 × 512 · f64 | 8036900 | 8036.9 |
| **Cholesky factorization / NumPy**<br>512 × 512 · f64 | 6134100 | 6134.1 |
| **Cholesky factorization / faer**<br>512 × 512 · f64 | 3098600 | 3098.6 |
| **Cholesky factorization / Kernix**<br>512 × 512 · f64 | 4684900 | 4684.9 |

# Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Column-pivoted QR / faer**<br>96 × 32 · f64 | 68700 | 68.7 |
| **Column-pivoted QR / nalgebra**<br>96 × 32 · f64 | 107900 | 107.9 |
| **Column-pivoted QR / Kernix**<br>96 × 32 · f64 | 118900 | 118.9 |
| **Column-pivoted QR / nalgebra**<br>256 × 64 · f64 | 925900 | 925.9 |
| **Column-pivoted QR / faer**<br>256 × 64 · f64 | 397600 | 397.6 |
| **Column-pivoted QR / Kernix**<br>256 × 64 · f64 | 886700 | 886.7 |
| **Minimum-norm least squares / nalgebra**<br>64 × 32, rank 16, 4 RHS · f64 | 104900 | 104.9 |
| **Minimum-norm least squares / NumPy**<br>64 × 32, rank 16, 4 RHS · f64 | 112300 | 112.3 |
| **Minimum-norm least squares / Kernix**<br>64 × 32, rank 16, 4 RHS · f64 | 88500 | 88.5 |
| **Minimum-norm least squares / NumPy**<br>32 × 64, rank 32, 4 RHS · f64 | 204800 | 204.8 |

# Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Minimum-norm least squares / Kernix**<br>32 × 64, rank 32, 4 RHS · f64 | 116900 | 116.9 |
| **Minimum-norm least squares / nalgebra**<br>32 × 64, rank 32, 4 RHS · f64 | 164300 | 164.3 |
| **Thin QR / faer**<br>96 × 32 · f64 | 86600 | 86.6 |
| **Thin QR / NumPy**<br>96 × 32 · f64 | 77900 | 77.9 |
| **Thin QR / Kernix**<br>96 × 32 · f64 | 101500 | 101.5 |
| **Thin QR / nalgebra**<br>96 × 32 · f64 | 82800 | 82.8 |
| **Thin QR / faer**<br>256 × 64 · f64 | 406200 | 406.2 |
| **Thin QR / nalgebra**<br>256 × 64 · f64 | 641300 | 641.3 |
| **Thin QR / Kernix**<br>256 × 64 · f64 | 757100 | 757.1 |
| **Thin QR / NumPy**<br>256 × 64 · f64 | 545800 | 545.8 |

# Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Cholesky factorization / Kernix**<br>64 × 64 · f64 | 36000 | 36 |
| **Cholesky factorization / faer**<br>64 × 64 · f64 | 14600 | 14.6 |
| **Cholesky factorization / NumPy**<br>64 × 64 · f64 | 21100 | 21.1 |
| **Cholesky factorization / nalgebra**<br>64 × 64 · f64 | 19700 | 19.7 |
| **Cholesky factorization / NumPy**<br>128 × 128 · f64 | 115100 | 115.1 |
| **Cholesky factorization / nalgebra**<br>128 × 128 · f64 | 112400 | 112.4 |
| **Cholesky factorization / faer**<br>128 × 128 · f64 | 59400 | 59.4 |
| **Cholesky factorization / Kernix**<br>128 × 128 · f64 | 223600 | 223.6 |
| **Cholesky factorization / NumPy**<br>256 × 256 · f64 | 943400 | 943.4 |
| **Cholesky factorization / faer**<br>256 × 256 · f64 | 314100 | 314.1 |

# Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Cholesky factorization / nalgebra**<br>256 × 256 · f64 | 863400 | 863.4 |
| **Cholesky factorization / Kernix**<br>256 × 256 · f64 | 699400 | 699.4 |
| **Shared-variable backward / Candle**<br>sum(X * X), 64 × 32; prepared graph · f64 | 50100 | 50.1 |
| **Shared-variable backward / Kernix**<br>sum(X * X), 64 × 32; prepared graph · f64 | 9800 | 9.8 |
| **Record + backward / Candle**<br>sum(A · W), 32 × 16 · 16 × 8 · f64 | 9800 | 9.8 |
| **Record + backward / Kernix**<br>sum(A · W), 32 × 16 · 16 × 8 · f64 | 4900 | 4.9 |
| **Model inference / Candle**<br>batched normalized/sliced MLP · f64 | 8200 | 8.2 |
| **Model inference / Kernix**<br>batched normalized/sliced MLP · f64 | 3500 | 3.5 |
| **Model recording + forward / Kernix**<br>batched normalized/sliced MLP · f64 | 4900 | 4.9 |
| **Model recording + forward / Candle**<br>batched normalized/sliced MLP · f64 | 10200 | 10.2 |

# Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Model backward / Kernix**<br>prepared graph → fresh gradients · f64 | 5900 | 5.9 |
| **Model backward / Candle**<br>prepared graph → fresh gradients · f64 | 31699.999999999996 | 31.699999999999996 |
| **Parameter update / Kernix**<br>transactional SGD, fixed gradients · f64 | 1000 | 1 |
| **Coalesced tensor contraction / Kernix**<br>16 × 32 × 8 · 32 × 8 × 16 · f64 | 63700 | 63.7 |
| **Coalesced tensor contraction / NumPy**<br>16 × 32 × 8 · 32 × 8 × 16 · f64 | 12200 | 12.2 |
| **Row softmax / Candle**<br>256 × 64 · f64 | 181800 | 181.8 |
| **Row softmax / ndarray**<br>256 × 64 · f64 | 153700 | 153.7 |
| **Row softmax / Kernix**<br>256 × 64 · f64 | 114200 | 114.2 |
| **Row softmax / NumPy**<br>256 × 64 · f64 | 133100 | 133.1 |
| **Create zeros / faer**<br>64 × 64 · f32 | 500 | 0.5 |

# Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Create zeros / Candle**<br>64 × 64 · f32 | 500 | 0.5 |
| **Create zeros / NumPy**<br>64 × 64 · f32 | 800 | 0.8 |
| **Create zeros / Kernix**<br>64 × 64 · f32 | 400 | 0.4 |
| **Create zeros / ndarray**<br>64 × 64 · f32 | 300 | 0.3 |
| **Create ones / Candle**<br>64 × 64 · f32 | 700 | 0.7 |
| **Create ones / ndarray**<br>64 × 64 · f32 | 500 | 0.5 |
| **Create ones / NumPy**<br>64 × 64 · f32 | 2100 | 2.1 |
| **Create ones / Kernix**<br>64 × 64 · f32 | 500 | 0.5 |
| **Create ones / faer**<br>64 × 64 · f32 | 500 | 0.5 |